

2023 - Year in Space

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In the last few years, we have seen some of the most important breakthroughs in the world of space science and technology. While there were tons of new discoveries in the last year, we also reaped the fruits of the seeds we sowed in 2022 and before. To celebrate our success in the world of space exploration and research, I bring you a collection of some of the biggest breakthroughs in space science from 2023. If you feel that I missed out on some event, then feel free to write to us.

Off into space we go!

In 2023, we also saw many people attempt to commercialise space travel and tourism in their own unique ways. We had SpaceX successfully send the Axiom Mission 2 to the International Space Station (ISS) on May 21, 2023. The mission was a success as the private crew that was sent to outer space returned safely to our pale blue dot on May 31, 2023. Four days after the launch of the Axiom Mission 2, we



saw Virgin Galactic launch the Unity 25 mission. It was a sub-orbital space flight with a six-person crew on board SpaceShipTwo spaceplane VSS Unity. Subsequently, there were other manned flights as well.

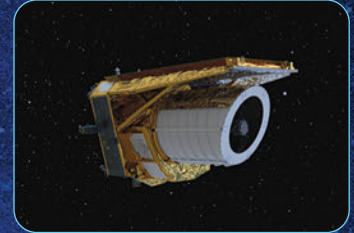
Riding high on success

During the course of the year, we saw only 11 out of the 200 missions fail. That puts it at a 0.05 per cent rate of failure. Impressive, isn't it!?

Month	Num. of successes	Num. of failures	Num. of partial failures
January	14	2	0
February	12	0	0
March	22	3	0
April	11	1	0
May	19	1	0
June	13	0	0
July	18	0	0
August	21	1	0
September	21	2	0
October	17	0	0
November	15	1	0
December	6	0	0
Total	189	11	0

Solar escapades

In 2023, the Sun seemed to be a major point of interest for space agencies as we saw not one, but two of the most active space agencies of the world,



ESA (European Space Agency) and ISRO (Indian Space Research Organisation) send missions to the Sun.

ESA's Euclid satellite, launched on July 1 was aimed at Sun-Earth L2 point on board a Falcon 9 rocket. The aim was to better understand two of the most complex subjects in the world of astronomy - dark matter and dark energy. This is to be done by further studying the acceleration of the universe. During its six years of expected mission duration, Euclid is expected to observe about 15,000 deg² (4.6 sr), about a third of the sky, with the primary focus being on the side that faces away from the Milky Way.

ISRO's Aditya-L1 was launched a couple of months after Euclid, on September 2, 2023, on board the PSLV C57 rocket. As stated by ISRO in their mission statement, the main objectives of the mission is to - "Study of Solar upper atmospheric (chromosphere and corona) dynamics," and more that adds on.!